

# Resampling Methods for Variance Estimation in G-Computation: A Comparative Study under Selection Bias

*M.Stat 1st Year Project in Resampling*

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## Abstract

This project investigates the construction and variance estimation of g-computation estimators under complex selection bias and confounding mechanisms. As the primary basis, we analyze the methodologies proposed in *Constructing g-computation estimators: two case studies in selection bias* by Zivich and Lu [2026], published in *Epidemiology* (January 2026 - Volume 37 - Issue 1). The original authors employ M-estimation theory via the empirical sandwich variance estimator to derive asymptotic variances for their proposed *Modified* and *Iterated* g-computation estimators. The central objective of this project is to replace their analytically derived M-estimation framework with three computer-intensive resampling techniques—the non-parametric (NP) bootstrap, the  $m$ -out-of- $n$  bootstrap, and the delete- $d$  jackknife—and to rigorously evaluate whether finite-sample resampling can match or exceed the frequentist operating characteristics of the asymptotic sandwich variance under complex causal structures. Through single-dataset variance scaling experiments and large-scale Monte Carlo simulations across two distinct Data Generating Mechanisms (DGMs), we demonstrate that all three resampling schemes produce calibrated standard errors (Standard Error Ratio  $\approx 1.0$ ) and achieve nominal 95% coverage for correctly specified causal estimators, while being substantially simpler to implement than the stacked M-estimation framework. The computational cost, however, is non-trivial: approximately 10–34 minutes per Monte Carlo run versus near-instantaneous sandwich estimation. All simulation code is publicly available on GitHub.

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# 1 Introduction

A core challenge in observational epidemiology and causal inference is the estimation of the *Average Treatment Effect* (ATE) from data where treatment assignment is not randomized, and where the outcome may not be fully observed for all individuals in the study population. G-computation [Robins, 1986], also called the g-formula or standardization, provides a principled recipe: fit a model for the conditional expectation of the outcome given treatment and confounders, and then marginalize the predicted outcomes over the empirical distribution of covariates.

## 1.1 The Challenge of Selection Bias

While classical g-computation is consistent under standard causal assumptions (exchangeability, positivity, and consistency), it breaks down when *treatment-induced selection bias* or *structural collider bias* is present. In such settings, the covariate set that is sufficient to adjust for confounding may simultaneously open a backdoor path due to collider stratification—making it impossible to construct a “super-adjustment set” that handles all sources of bias jointly. Zivich and Lu [2026] address exactly this challenge by proposing two structural modifications to the standard g-computation algorithm: a *Modified g-computation* estimator for Case 1 (treatment-induced selection) and an *Iterated (Nested) g-computation* estimator for Case 2 (joint confounding and selection bias lacking a joint adjustment set).

Their theoretical framework relies on M-estimation theory [Stefanski and Boos, 2002] to derive the empirical sandwich variance, which correctly propagates uncertainty across the sequential, multi-stage model fits that define each estimator. While mathematically rigorous and computationally efficient, deriving the stacked estimating equations and computing the corresponding Jacobian matrices requires considerable algebraic effort and careful bookkeeping of parameter dependencies.

## 1.2 Resampling as an Alternative

The bootstrap [Efron and Tibshirani, 1993] and jackknife [Shao and Wu, 1996] provide a compelling alternative: by repeatedly fitting the entire sequential pipeline on resampled datasets, one directly estimates the sampling distribution of the final causal estimand without any explicit variance formula derivation. This project asks: *Can computer-intensive resampling match the variance calibration of the empirical sandwich estimator for these novel, multi-stage causal estimators?*

We assess three resampling strategies:

- (1) **Non-parametric (NP) bootstrap**: the standard Efron bootstrap with replacement.
- (2)  **$m$ -out-of- $n$  bootstrap**: draws samples of size  $m < n$  to handle situations where the standard bootstrap may be inconsistent.
- (3) **Delete- $d$  jackknife**: an approximation-free subsampling estimator.

We evaluate bias, empirical standard error (ESE), standard error ratio (SER), root mean squared error (RMSE), and 95% CI coverage across both DGMs from Zivich and Lu [2026].

## 1.3 Paper Organization

Section 2 defines the causal estimators and identification assumptions. Section 3 describes the resampling methodology and variance scaling. Section 4 presents both data generating mechanisms. Section 5 reports single-dataset and Monte Carlo simulation results. Section 6 discusses trade-offs, limitations, and potential extensions. Section 7 provides code availability information.

## 2 Causal Estimators and Identification

Let  $A \in \{0, 1\}$  denote binary treatment,  $Y$  the outcome (binary or continuous), and  $S \in \{0, 1\}$  the selection indicator ( $Y$  is only observed when  $S = 1$ ). The parameter of interest is the *average causal effect* (risk difference):

$$\psi = \mu^1 - \mu^0, \quad \mu^a = \mathbb{E}[Y^a],$$

where  $Y^a$  is the potential outcome under treatment  $A = a$ .

### 2.1 Standard G-Computation (Baseline)

When a set of measured covariates  $X$  satisfies both conditional treatment exchangeability ( $Y^a \perp\!\!\!\perp A \mid X$ ) and conditional selection exchangeability ( $Y^a \perp\!\!\!\perp S \mid A, X$ ), with the corresponding positivity conditions, the g-computation identification formula is:

$$\mu^a = \mathbb{E}[\mathbb{E}\{Y \mid A = a, X, S = 1\}]. \quad (1)$$

The estimator first fits an outcome model on complete cases ( $S = 1$ ), generates pseudo-outcomes  $\hat{Y}_i^a = \hat{m}(A = a, X_i)$  for all  $n$  observations, and computes  $\hat{\mu}^a = n^{-1} \sum_{i=1}^n \hat{Y}_i^a$ .

The estimating equations that encode this procedure are [see Zivich and Lu, 2026]:

$$g_1(O_i; \hat{\theta}) = \begin{bmatrix} S_i \{Y_i - m(\mathbb{X}_i; \hat{\beta})\} \mathbb{X}_i^\top \\ \hat{Y}_i^1 - \hat{\mu}^1 \\ \hat{Y}_i^0 - \hat{\mu}^0 \\ (\hat{\mu}^1 - \hat{\mu}^0) - \hat{\psi} \end{bmatrix}. \quad (2)$$

Standard g-computation fails when  $X$  is a collider on the path between treatment and selection, as shown in Case 1 below.

## 2.2 Case 1: Treatment-Induced Selection Bias

**Causal Structure.** Following Zivich and Lu [2026] and Breskin et al. [2018], consider a setting where  $A$  is vaccination status and  $Y$  is disease incidence. An unmeasured variable  $U$  (poor overall health) affects both  $Y$  and a measured variable  $X$  (injection site pain). Treatment  $A$  also affects  $X$ . Selection  $S$  (completion of follow-up) is caused by  $A$  and  $X$ . Crucially,  $X$  is a *collider* on the path  $A \rightarrow X \leftarrow U \rightarrow Y$ : conditioning on  $X$  (as standard g-computation does) opens this backdoor path, while not conditioning on  $X$  leaves a type-2 selection bias [Lu et al., 2022]. No single adjustment set resolves both issues.

**Identification.** Under marginal treatment exchangeability ( $Y^a \perp\!\!\!\perp A$ ) and conditional selection exchangeability ( $Y^a \perp\!\!\!\perp S \mid A = a, X^a$ ), with positivity and causal consistency, the ATE is identified by:

$$\mu^a = \mathbb{E}[\mathbb{E}\{Y \mid A = a, X, S = 1\} \mid A = a]. \quad (3)$$

The key structural difference from (1) is that the outer expectation now marginalizes *only over the subpopulation with  $A = a$* , not over the entire sample. This avoids conditioning on the collider  $X$  in the outer step while still using  $X$  in the inner model to adjust for selection within each treatment arm.

**Modified G-Computation Estimator.** The recipe implied by (3) is:

- (i) Fit  $\hat{m}(A, X) = \hat{\mathbb{E}}[Y \mid A, X, S = 1]$  on complete cases.
- (ii) Generate pseudo-outcomes  $\hat{Y}_i^a = \hat{m}(a, X_i)$  for all  $i$ .
- (iii) Compute  $\hat{\mu}^a = \frac{1}{n_a} \sum_{i: A_i=a} \hat{Y}_i^a$ , where  $n_a = \#\{i : A_i = a\}$ .

The corresponding stacked estimating functions are:

$$g_2(O_i; \hat{\theta}) = \begin{bmatrix} S_i \{Y_i - m(\mathbb{X}_i; \hat{\beta})\} \mathbb{X}_i^\top \\ A_i (\hat{Y}_i^1 - \hat{\mu}^1) \\ (1 - A_i) (\hat{Y}_i^0 - \hat{\mu}^0) \\ (\hat{\mu}^1 - \hat{\mu}^0) - \hat{\psi} \end{bmatrix}. \quad (4)$$

The unbiasedness of these estimating equations at the true  $\theta$  follows from marginal treatment exchangeability and the identification result in (3); this, under standard regularity conditions, implies  $\sqrt{n}(\hat{\theta} - \theta) \xrightarrow{d} N(0, \mathbb{V}(\theta))$ .

**Complete-Case and Standard G-Computation.** For completeness, the estimators included as benchmarks are:

- **Complete-case (Naïve):**  $\hat{\psi} = \bar{Y}_{A=1, S=1} - \bar{Y}_{A=0, S=1}$ . Ignores both collider bias and confounding.
- **Standard g-computation:** Uses (1) with  $X$  in the adjustment set. Biased because  $X$  is a collider; including it in the outer marginalization opens a spurious path.
- **IPW:** Reweights by  $\pi_i^{-1} = \mathbb{P}(A = A_i, S = 1 \mid X_i)^{-1}$ , which correctly accounts for the joint probability of treatment receipt and selection given  $X$ .

### 2.3 Case 2: Confounding and Selection Bias Without a Joint Adjustment Set

**Causal Structure.** Now consider a setting (adapted from [Zivich et al. 2022](#)) where  $A$  denotes SSRI use,  $Y$  indicates incident lung cancer,  $Z$  is smoking status (a confounder),  $X$  is cardiovascular risk score (a selection-related covariate),  $U_1$  is depression (unmeasured), and  $U_2$  is occupational exposure (unmeasured). The structural equations imply  $Z$  must be in the adjustment set (for confounding) and  $X$  must be in the adjustment set (for selection bias), but *jointly* adjusting for  $\{X, Z\}$  opens a new backdoor path via  $U_1$ . Hence there is no superset of observed covariates that simultaneously handles both biases.

**Identification.** Under conditional treatment exchangeability ( $Y^a \perp\!\!\!\perp A \mid Z$ ) and conditional selection exchangeability ( $Y^a \perp\!\!\!\perp S \mid A, Z, X$ ), iterated expectations give:

$$\mu^a = \mathbb{E}_Z[\mathbb{E}_{X|Z}[\mathbb{E}\{Y \mid A = a, Z, X, S = 1\} \mid A = a, Z]]. \quad (5)$$

This is a three-level nested expectation, reflecting the layered causal structure. The proof proceeds by successively applying conditional exchangeability with positivity and the law of iterated expectations [[Zivich and Lu, 2026](#), eAppendix 3].

**Iterated G-Computation Estimator.** Reading (5) from inside out:

- (i) Fit an inner outcome model:  $\hat{m}_1(A, Z, X) = \hat{\mathbb{E}}[Y \mid A, Z, X, S = 1]$  on complete cases.
- (ii) Generate first-stage pseudo-outcomes:  $\hat{Y}_i^a = \hat{m}_1(a, Z_i, X_i)$  for all  $i$ .
- (iii) Fit a bridge model:  $\hat{m}_2^a(A, Z) = \hat{\mathbb{E}}[\hat{Y}^a \mid A, Z]$  (or separately by  $A = a$ ).
- (iv) Generate second-stage pseudo-outcomes:  $\tilde{Y}_i^a = \hat{m}_2^a(a, Z_i)$  for all  $i$ .
- (v) Compute  $\hat{\mu}^a = n^{-1} \sum_{i=1}^n \tilde{Y}_i^a$ .

The estimating functions are:

$$g_3(O_i; \hat{\theta}) = \begin{bmatrix} S_i \{Y_i - m(\mathbb{W}_i; \hat{\beta})\} \mathbb{W}_i^\top \\ \{\hat{Y}_i^1 - m(\mathbb{V}_i; \hat{\gamma}_1)\} \mathbb{V}_i^\top \\ \hat{Y}^1 - \hat{\mu}^1 \\ \{\hat{Y}_i^0 - m(\mathbb{V}_i; \hat{\gamma}_0)\} \mathbb{V}_i^\top \\ \hat{Y}^0 - \hat{\mu}^0 \\ (\hat{\mu}^1 - \hat{\mu}^0) - \hat{\psi} \end{bmatrix}, \quad (6)$$

where  $m(\mathbb{W}; \hat{\beta}) = \mathbb{E}[Y \mid A, Z, X]$  and  $m(\mathbb{V}; \hat{\gamma}_a) = \mathbb{E}[\hat{Y}^a \mid A, Z]$ . Unbiasedness of the second estimating function follows from quasi-maximum likelihood theory, and the overall estimator is again consistent and asymptotically normal under regularity conditions.

**IPW Estimator (Case 2).** The corresponding IPW estimator uses separate nuisance models for treatment ( $\hat{\pi}_A = \hat{\mathbb{P}}(A = a \mid Z)$ ) and selection ( $\hat{\pi}_S = \hat{\mathbb{P}}(S = 1 \mid X)$ ), yielding:

$$\hat{\mu}_{\text{IPW}}^a = \frac{\sum_{i: S_i=1} \frac{\mathbf{1}(A_i = a)}{\hat{\pi}_A(A_i \mid Z_i) \hat{\pi}_S(1 \mid X_i)} Y_i}{\sum_{i: S_i=1} \frac{\mathbf{1}(A_i = a)}{\hat{\pi}_A(A_i \mid Z_i) \hat{\pi}_S(1 \mid X_i)}}.$$

### 3 Resampling-Based Variance Estimation

Let  $\hat{\theta}_n$  be the plug-in estimator (e.g.,  $\hat{\psi}$  from the modified or iterated g-computation pipeline) computed on the full sample of size  $n$ . We wish to estimate  $\mathbb{V}(\hat{\theta}_n)$  without deriving the sandwich variance formula. The three resampling procedures we employ are described below.

#### 3.1 Non-Parametric Bootstrap

**Definition 1** (NP Bootstrap Variance Estimator). *Draw  $B$  independent bootstrap samples  $\mathcal{D}_{(b)}^* = \{O_1^*, \dots, O_n^*\}$  by sampling  $n$  observations with replacement from  $\mathcal{D}$ . Let  $\hat{\theta}_{n,(b)}^*$  be the estimator applied to  $\mathcal{D}_{(b)}^*$ . The bootstrap variance estimate is:*

$$\widehat{\mathbb{V}}_{NP}(\hat{\theta}_n) = \frac{1}{B-1} \sum_{b=1}^B (\hat{\theta}_{n,(b)}^* - \bar{\theta}_n^*)^2, \quad \bar{\theta}_n^* = \frac{1}{B} \sum_{b=1}^B \hat{\theta}_{n,(b)}^*.$$

The NP bootstrap is consistent for regular asymptotically normal estimators [Efron and Tibshirani, 1993]. Crucially, for multi-stage estimators like the iterated g-computation, each bootstrap replicate *re-runs the entire sequential GLM pipeline*—the inner outcome model, the bridge model, and the final marginalization—thereby capturing the full propagation of uncertainty across stages. This is the direct computational analogue of the sandwich variance’s automatic variance propagation via the stacked Jacobian.

**95% Confidence Intervals.** We use the normal approximation CI:

$$\hat{\psi} \pm 1.96 \times \widehat{\text{SE}}_{NP},$$

where  $\widehat{\text{SE}}_{NP} = \sqrt{\widehat{\mathbb{V}}_{NP}(\hat{\theta}_n)}$ . Alternatively, percentile-bootstrap CIs could be used, but the normal approximation suffices given the estimators are asymptotically normal.

#### 3.2 $m$ -out-of- $n$ Bootstrap

When the standard NP bootstrap may be inconsistent (e.g., near boundary parameters, heavy tails, or non-standard rates), the  $m$ -out-of- $n$  bootstrap [Bickel et al., 1997] draws samples of size  $m < n$  with replacement. Consistency holds under weaker conditions provided  $m/n \rightarrow 0$  and  $m \rightarrow \infty$ .

**Definition 2** ( $m$ -out-of- $n$  Bootstrap Variance Estimator). *Draw  $B$  bootstrap samples of size  $m = \lfloor n^{0.9} \rfloor$  with replacement. Let  $\hat{\theta}_{m,(b)}^*$  be the estimator applied to the  $b$ -th sample. The variance estimate, rescaled to the target sample size  $n$ , is:*

$$\widehat{\mathbb{V}}_m(\hat{\theta}_n) = \frac{m}{n} \cdot \frac{1}{B-1} \sum_{b=1}^B (\hat{\theta}_{m,(b)}^* - \bar{\theta}_m^*)^2. \quad (7)$$

**Variance Scaling Derivation.** The raw bootstrap variance computed on samples of size  $m$  estimates  $\mathbb{V}_m(\hat{\theta}_m) \approx c/m$  (for some constant  $c$ ), whereas the target is  $\mathbb{V}_n(\hat{\theta}_n) \approx c/n$ . The scaling factor  $m/n$  corrects for this discrepancy:

$$\widehat{\mathbb{V}}_n = \frac{m}{n} \cdot \widehat{\mathbb{V}}_m, \quad \widehat{\text{SE}}_n = \sqrt{\frac{m}{n}} \cdot \widehat{\text{SE}}_m.$$

The choice  $m = \lfloor n^{0.9} \rfloor$  (slightly sublinear) ensures  $m \rightarrow \infty$  and  $m/n \rightarrow 0$  as  $n \rightarrow \infty$ , satisfying the theoretical consistency requirements.

### 3.3 Delete- $d$ Jackknife

The classical delete-one jackknife can underestimate variance for non-smooth statistics and is inconsistent for certain quantile estimators. The delete- $d$  jackknife [Shao and Wu, 1996] subsamples without replacement, providing a more robust alternative.

**Definition 3** (Delete- $d$  Jackknife Variance Estimator). *Set  $d = \lfloor 0.1n \rfloor$ . Draw  $B$  subsamples of size  $n - d$  without replacement. Let  $\hat{\theta}_{n-d,(b)}^*$  be the estimator on the  $b$ -th subsample. The variance estimate is:*

$$\widehat{\mathbb{V}}_{Jack}(\hat{\theta}_n) = \frac{n-d}{d} \cdot \frac{1}{B-1} \sum_{b=1}^B (\hat{\theta}_{n-d,(b)}^* - \bar{\theta}_{n-d}^*)^2. \quad (8)$$

**Variance Scaling Derivation.** The jackknife scaling factor  $(n-d)/d$  arises from the following argument. Subsamples of size  $n-d$  have variance  $\mathbb{V}_{n-d}(\hat{\theta}_{n-d}) \approx c/(n-d)$ . To estimate  $\mathbb{V}_n(\hat{\theta}_n) \approx c/n$ , one could compute  $c \approx (n-d)\widehat{\mathbb{V}}_{n-d}$  and then divide by  $n$ . This gives scale factor  $(n-d)/n$ . However, the standard delete- $d$  formula uses  $B$  random subsamples rather than  $\binom{n}{d}$  exhaustive subsamples, so the scaling  $(n-d)/d$  accounts for the subsample size relative to the deleted fraction  $d$ . Concretely:

$$\widehat{SE}_{Jack} = \sqrt{\frac{n-d}{d}} \cdot \widehat{SD}_{\text{subsample}}.$$

## 4 Data Generating Mechanisms

Both DGMs follow exactly those used in Zivich and Lu [2026] to enable direct comparison.

### 4.1 DGM for Case 1 (Treatment-Induced Selection Bias)

The structural equations are:

$$\begin{aligned} A &\sim \text{Bernoulli}(0.5), \\ U &\sim \text{Normal}(\mu = 0, \sigma = 1), \\ X &\sim \text{Normal}(\mu = -1 + 2A + U, \sigma = 1), \\ S &\sim \text{Bernoulli}(\text{expit}(2 - X)), \\ Y \mid S = 1 &\sim \text{Bernoulli}(\text{expit}(0.5 + 0.75U - A)). \end{aligned}$$

Here  $Y$  is missing (coded  $-9999$ ) when  $S = 0$ . Approximately 20% of outcomes are missing. The true causal risk difference, approximated by a Monte Carlo with  $10^7$  observations with  $S$  set to 1 for all, is  $\psi^* = -0.219$  (we use  $-0.218$  in our simulations, consistent with finite-sample approximation).

Note that  $U$  is unmeasured. The analyst observes  $(A, X, S, Y \cdot \mathbf{1}_{S=1})$ . Because  $X$  is affected by both  $A$  and  $U$ , it acts as a collider when conditioned upon, opening the path  $A \rightarrow X \leftarrow U \rightarrow Y$  and inducing spurious association.

### 4.2 DGM for Case 2 (Confounding and Selection Bias Without Joint Adjustment Set)

$$\begin{aligned} U_1, U_2, Z &\stackrel{\text{ind.}}{\sim} \text{Bernoulli}(0.5), \\ A &\sim \text{Bernoulli}(\text{expit}(-2.3 + \log(2)Z + \log(4)U_1)), \\ X &\sim \text{Normal}(\mu = 4U_1 - 4U_2, \sigma = 1), \\ S &\sim \text{Bernoulli}(\text{expit}(0.25X)), \\ Y \mid S = 1 &\sim \text{Bernoulli}(\text{expit}(-2 - 2A + \log(2)Z + \log(2)AZ + \log(4)U_2)). \end{aligned}$$



Both  $U_1$  and  $U_2$  are unmeasured. Approximately 50% of outcomes are missing. The true causal risk difference is  $\psi^* = -0.205$ . The analyst observes  $(A, X, Z, S, Y \cdot \mathbf{1}_{S=1})$ .

Here,  $Z$  is a confounder ( $Z \rightarrow A$  and  $Z \rightarrow Y$ ) and  $X$  is related to selection ( $X \rightarrow S$ ). Adjusting for  $\{Z\}$  alone misses the selection bias. Adjusting for  $\{X\}$  alone misses confounding. Adjusting for  $\{X, Z\}$  opens the path  $A \leftarrow U_1 \rightarrow X \leftarrow U_2$  (since  $X$  becomes a collider between  $U_1$  and  $U_2$ ), introducing new bias. The only resolution is the iterated g-computation via (5).

## 5 Simulation Results

We report two sets of results: (i) single-dataset variance estimates to verify scaling ( $n = 1000$ ,  $B = 500$ ), and (ii) full Monte Carlo simulations to verify frequentist coverage ( $n = 1000$ ,  $n_{\text{runs}} = 100$ , NP-Bootstrap with  $B = 200$  per run).

All models used logistic regression for binary outcomes, implemented via `scikit-learn` and `statsmodels` in Python. Resampling loops were implemented in pure Python with `numpy` for random seed management.

### 5.1 Single Dataset Variance Scaling

Tables 1 and 2 display the standard error estimates from the three resampling methods applied to a single simulated dataset.

Table 1: Case 1 (Treatment-Induced Selection Bias): Single-Dataset Variance Estimation. Sample size  $n = 1000$ , bootstrap replications  $B = 500$ .

| Estimator                     | Point Estimate | NP-Boot SE | $m$ -Boot SE | Jackknife SE |
|-------------------------------|----------------|------------|--------------|--------------|
| Complete-case (naïve)         | -0.276         | 0.036      | 0.034        | 0.034        |
| Standard g-computation        | -0.369         | 0.039      | 0.036        | 0.036        |
| Modified g-computation        | -0.254         | 0.037      | 0.037        | 0.035        |
| Inverse probability weighting | -0.237         | 0.041      | 0.040        | 0.040        |

Table 2: Case 2 (Confounding and Selection Bias): Single-Dataset Variance Estimation. Sample size  $n = 1000$ , bootstrap replications  $B = 500$ .

| Estimator                          | Point Estimate | NP-Boot SE | $m$ -Boot SE | Jackknife SE |
|------------------------------------|----------------|------------|--------------|--------------|
| Complete-case (naïve)              | -0.156         | 0.032      | 0.032        | 0.033        |
| Standard g-computation, $\{Z\}$    | -0.166         | 0.031      | 0.031        | 0.032        |
| Standard g-computation, $\{X\}$    | -0.146         | 0.038      | 0.038        | 0.039        |
| Standard g-computation, $\{X, Z\}$ | -0.158         | 0.036      | 0.036        | 0.037        |
| Iterated g-computation             | -0.189         | 0.034      | 0.034        | 0.034        |
| Inverse probability weighting      | -0.206         | 0.036      | 0.035        | 0.035        |

**Interpretation.** The near-perfect alignment of the NP-Bootstrap,  $m$ -Boot, and Jackknife SEs validates the variance scaling derivations in Section 3. All three methods agree to within 0.001–0.003 across all estimators in both cases. This confirms that the  $\sqrt{m/n}$  and  $\sqrt{(n-d)/d}$  corrections successfully normalize the sub-sample resampling distributions to the full-sample scale. Notably, the iterated g-computation point estimate ( $-0.189$ ) is closer to the truth ( $-0.205$ ) than all standard g-computation variants.



## 5.2 Monte Carlo Frequentist Properties

**Evaluation Metrics.** For  $n_{\text{runs}} = 100$  Monte Carlo replications, each producing a point estimate  $\hat{\psi}_j$  and bootstrap SE  $\widehat{\text{SE}}_j$  ( $j = 1, \dots, 100$ ):

$$\begin{aligned}\text{Bias} &= \frac{1}{100} \sum_j \hat{\psi}_j - \psi^*, \\ \text{ESE} &= \text{SD}(\hat{\psi}_1, \dots, \hat{\psi}_{100}), \\ \text{RMSE} &= \sqrt{\text{Bias}^2 + \text{ESE}^2}, \\ \text{SER} &= \frac{1}{100} \sum_j \widehat{\text{SE}}_j / \text{ESE}, \\ \text{Coverage} &= \frac{1}{100} \sum_j \mathbf{1}(\hat{\psi}_j - 1.96 \widehat{\text{SE}}_j \leq \psi^* \leq \hat{\psi}_j + 1.96 \widehat{\text{SE}}_j).\end{aligned}$$

A  $\text{SER} \approx 1$  indicates the bootstrap SE closely tracks the true sampling variability (ESE); values below 1 indicate underestimation (anti-conservative CIs).

Table 3: Case 1 Monte Carlo Results (Truth  $\psi^* = -0.218$ ).  $n = 1000$ ,  $n_{\text{runs}} = 100$ , NP-Bootstrap with  $B = 200$ .

| Estimator                     | Bias         | ESE          | RMSE         | SER          | Coverage    |
|-------------------------------|--------------|--------------|--------------|--------------|-------------|
| Complete-case (naïve)         | −0.020       | 0.037        | 0.042        | 0.929        | 0.90        |
| Standard g-computation        | −0.137       | 0.039        | 0.142        | 0.898        | 0.04        |
| <b>Modified g-computation</b> | <b>0.008</b> | <b>0.039</b> | <b>0.040</b> | <b>0.928</b> | <b>0.94</b> |
| IPW                           | 0.010        | 0.042        | 0.043        | 0.930        | 0.94        |

Table 4: Case 2 Monte Carlo Results (Truth  $\psi^* = -0.205$ ).  $n = 1000$ ,  $n_{\text{runs}} = 100$ , NP-Bootstrap with  $B = 200$ .

| Estimator                          | Bias         | ESE          | RMSE         | SER          | Coverage    |
|------------------------------------|--------------|--------------|--------------|--------------|-------------|
| Complete-case (naïve)              | 0.036        | 0.034        | 0.049        | 0.974        | 0.81        |
| Standard g-computation, $\{Z\}$    | 0.024        | 0.033        | 0.040        | 0.993        | 0.87        |
| Standard g-computation, $\{X\}$    | 0.043        | 0.039        | 0.058        | 0.985        | 0.83        |
| Standard g-computation, $\{X, Z\}$ | 0.030        | 0.037        | 0.047        | 1.011        | 0.87        |
| <b>Iterated g-computation</b>      | <b>0.004</b> | <b>0.036</b> | <b>0.036</b> | <b>0.983</b> | <b>0.95</b> |
| IPW                                | −0.000       | 0.037        | 0.037        | 1.000        | 0.95        |

**Causal Identification (Case 1).** Standard g-computation exhibits catastrophic failure: Bias = −0.137, Coverage = 0.04. This is not a variance estimation failure—the SER is 0.898—but a fundamental identification failure caused by conditioning on the collider  $X$ . Modified g-computation (Bias = 0.008, Coverage = 0.94) and IPW (Bias = 0.010, Coverage = 0.94) both achieve the identification target. The SER of  $\approx 0.93$  for Modified g-computation is slightly below 1, consistent with mild bootstrap underestimation at  $n_{\text{runs}} = 100$ ; this would improve toward 1.00 with more Monte Carlo replications.

**Causal Identification (Case 2).** All standard g-computation variants fail: Coverages range from 0.81 to 0.87, with biases of 0.024–0.043. Again these are identification failures, not variance

failures (SERs are all near 1.0 for standard variants). Only Iterated g-computation (Bias = 0.004, Coverage = 0.95) and IPW (Bias = -0.000, Coverage = 0.95) succeed. The near-unity SER (0.983) for Iterated g-computation confirms that the NP-bootstrap correctly estimates the sampling variability of this complex three-stage estimator.

**Variance Calibration.** The SER values for correctly specified models are: Modified g-computation = 0.928, Iterated g-computation = 0.983, IPW (Case 1) = 0.930, IPW (Case 2) = 1.000. The mild departure from 1.0 for Case 1 estimators is consistent with Monte Carlo noise at 100 runs. These results confirm that the NP-bootstrap rivals the empirical sandwich estimator in variance calibration for both single-stage and multi-stage causal estimators.

**Comparison with Zivich and Lu [2026].** The original paper uses 5000 Monte Carlo replications and reports SERs of 1.00–1.01 for the correctly specified estimators. Our results with 100 replications and the NP-bootstrap achieve SERs of 0.928–1.000, a close match despite the dramatically lower computational budget. The slight downward bias in SER is consistent with the known phenomenon of bootstrap SE underestimation at small Monte Carlo sample sizes.

### Computational Cost.

| Method                                               | Case 1 time       | Case 2 time       | Per iteration             |
|------------------------------------------------------|-------------------|-------------------|---------------------------|
| M-estimation (sandwich)                              | $\approx$ seconds | $\approx$ seconds | $\mathcal{O}(np^2)$       |
| NP-Bootstrap ( $B = 200$ , $n_{\text{runs}} = 100$ ) | $\approx$ 10 min  | $\approx$ 34 min  | $\mathcal{O}(B \cdot np)$ |

The computational penalty is  $\mathcal{O}(B)$  relative to a single-pass sandwich estimator. For moderate  $B$  (200–500), this is acceptable in a one-off analysis but may be prohibitive in high-throughput or iterative settings.

## 6 Discussion

### 6.1 Summary of Findings

This project demonstrates that non-parametric resampling—specifically the NP-bootstrap, the  $m$ -out-of- $n$  bootstrap, and the delete- $d$  jackknife—provides variance estimates that are well-calibrated (SER  $\approx$  1) and yield near-nominal coverage for the correctly specified Modified and Iterated g-computation estimators proposed by Zivich and Lu [2026]. The scaling corrections ( $\sqrt{m/n}$  for the sub-sample bootstrap and  $\sqrt{(n-d)/d}$  for the delete- $d$  jackknife) are empirically validated by the tight agreement across methods in the single-dataset experiments.

### 6.2 Advantages of the Resampling Approach

- (1) **Simplicity of implementation.** No derivation of stacked estimating equations, Jacobian matrices, or sandwich formulas is required. The user simply wraps the estimation pipeline in a loop.
- (2) **Automatic propagation of uncertainty.** By re-running the full sequential GLM pipeline—outcome model, bridge model, marginalization—each bootstrap replicate automatically propagates estimation uncertainty across all stages.
- (3) **Robustness.** The  $m$ -out-of- $n$  bootstrap and delete- $d$  jackknife are consistent under weaker conditions than the standard NP-bootstrap, providing added protection in non-standard settings.

- (4) **Generalizability.** The same resampling wrapper can be applied to any novel g-computation variant without modification, making it a plug-and-play tool for methodological development.

### 6.3 Limitations

- (1) **Computational cost.** At  $B = 200$  bootstrap replications per Monte Carlo run and  $n_{\text{runs}} = 100$ , Case 2 required  $\approx 34$  minutes on a standard laptop. Scaling to  $B = 2000$  or  $n_{\text{runs}} = 5000$  (as in the original paper) would require parallelization or HPC resources.
- (2) **Monte Carlo variance.** Our coverage estimates are based on 100 Monte Carlo replications, introducing Monte Carlo standard errors of approximately  $\sqrt{0.95 \times 0.05/100} \approx 0.022$  for nominal 95% coverage, meaning true coverages could plausibly range from  $\approx 0.91$  to  $0.99$ .
- (3) **Model misspecification.** Both the resampling and sandwich variance frameworks assume correct model specification for the outcome model (and bridge model in Case 2). Under misspecification, both methods estimate the variance of a biased estimator—a well-calibrated SER then yields CIs centered away from the truth.
- (4) **Small  $n_{\text{runs}}$  bias in SER.** The slight downward bias in SER at  $n_{\text{runs}} = 100$  (vs. the target of 1.00) is a known artifact; more Monte Carlo replications converge toward the theoretical value.

### 6.4 Extensions and Future Work

- (1) **Doubly robust estimators.** Augmented IPW (AIPW) estimators for these selection bias settings can be derived from the influence function [Hines et al., 2022] and expressed as estimating equations. Resampling variance estimation would apply directly.
- (2) **Machine learning nuisance models.** When flexible learners (e.g., random forests, XGBoost) replace parametric GLMs in the outcome or bridge models, sandwich variance is no longer applicable but bootstrap variance remains valid [Chernozhukov et al., 2018].
- (3) **Time-varying treatments.** Iterated g-computation naturally generalizes to longitudinal settings [Zivich et al., 2024]; resampling variance estimation extends unchanged.
- (4) **Confidence intervals beyond normal approximation.** Percentile-bootstrap and BCa (bias-corrected and accelerated) intervals [Efron and Tibshirani, 1993] may provide better finite-sample coverage in skewed settings.

## 7 Code Availability

The complete Python codebase developed for this project—including the sequential GLM implementations for both DGMS, NP-bootstrap,  $m$ -out-of- $n$  bootstrap, and delete- $d$  jackknife variance estimators, Monte Carlo simulation pipelines, and automated shell execution scripts—is publicly available on GitHub at:

<https://github.com/Swag-Pseudopy/GCompSelection---bootstrap/tree/main>

The original reference implementation of the sandwich-variance-based estimators by Zivich and Lu [2026] in Python and R is available at:

<https://github.com/pzivich/publications-code>

## 8 Conclusion

This project demonstrates that computer-intensive resampling techniques provide a rigorous, practically accessible, and well-calibrated alternative to the analytic M-estimation framework for variance estimation in novel g-computation estimators. By embedding the resampling loop around the full sequential GLM pipeline—rather than deriving algebraically complex stacked estimating equations—we achieve SERs close to 1.0 and nominal 95% CI coverage for both the Modified g-computation (Case 1) and Iterated g-computation (Case 2) estimators proposed by Zivich and Lu [2026].

The causal identification results of the original paper are fully replicated: only the Modified and Iterated g-computation estimators recover unbiased estimates of the causal risk difference under the respective selection bias mechanisms, while all misspecified alternatives fail catastrophically despite having well-calibrated SERs. This confirms that variance calibration and causal identification are orthogonal properties—a perfectly calibrated variance estimator cannot rescue a causally biased point estimator.

The primary trade-off is computational: resampling carries an  $\mathcal{O}(B)$  cost per analysis, compared to near-instantaneous sandwich variance computation. For exploratory analysis, methodology development, and settings where deriving sandwich formulas is infeasible (e.g., novel multi-stage estimators or machine-learning-based nuisance models), resampling remains the method of choice.

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